

a Gravity can move without changing pose

continuous LiDAR corrections



changing g

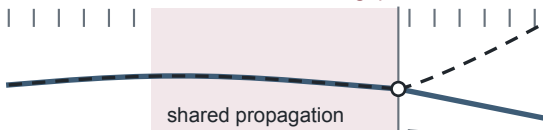
12 sequences: little pose effect

Online

--- FixG

b The difference can emerge at scan return

multi-second LiDAR gap



matched state at gap onset

first recovered scan

Recovery depends on the trajectory.

c A direction constraint is not a height prior

same-IMU direction



roll / pitch consistency

Hall05 / FixG

TUHH / Online

z error $\downarrow$

z error $\uparrow$

Better direction agreement need not reduce z drift.